УНИВЕРСИТЕТ ИТМО

Технологии машинного обучения
Machine learning technologies

Lecture 1 – Introduction

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ITMO University

From AI to AGI: state of 2026

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What AI can do in 2026?

- Creates websites/develops applications
- Personal tutors
- Conversational bots
- Writing/text generation
- Image/Video generation
- Autonomous AI agents for finance and industry
- Generate music, designs etc.,
- Scientific breakthroughs
- and many more…

The transition from narrow AI to Artificial General Intelligence (AGI) is widely debated in 2026

AGI - AI system that can match or exceed human cognitive abilities across any intellectual task

- Major labs declaring the AGI era has begun - e.g., Greg Brockman (OpenAI President), after release of OpenAI's GPT-6 Astra
- Independent researchers point to lingering gaps in reasoning, continuous learning, and real-world autonomy.

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"high scores on specific test benchmarks do not mean true general intelligence"

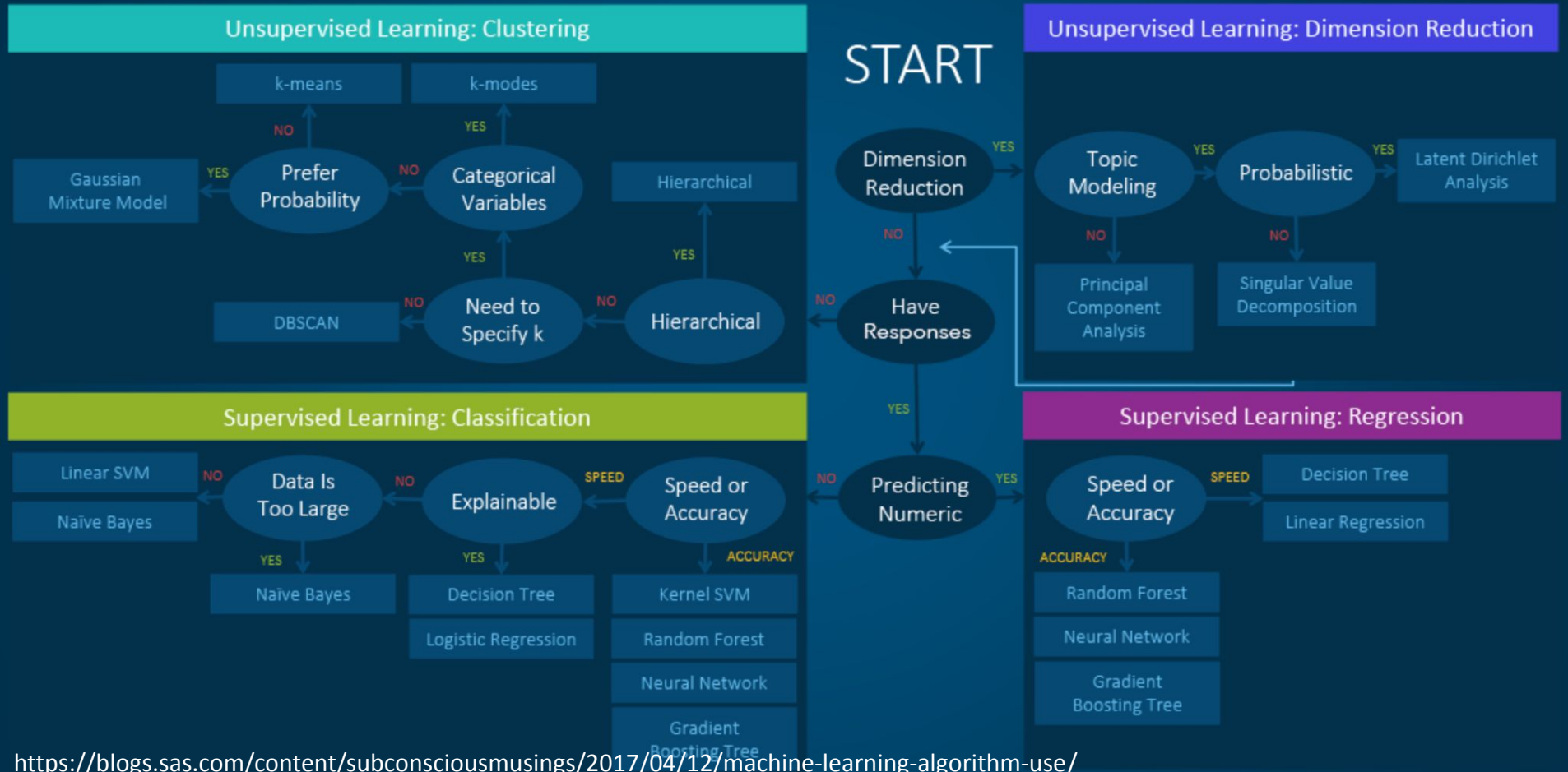
“Will Smith eats spaghetti”

Video generation benchmark (Text2Video)



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Machine Learning Algorithms Cheat Sheet



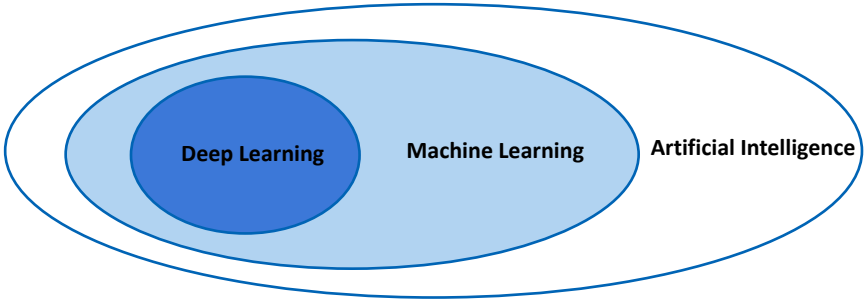
https://blogs.sas.com/content/subconsciousmusings/2017/04/12/machine-learning-algorithm-use/

Some basic definitions

What is Machine Learning?

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Machine learning -
A branch of artificial intelligence that studies methods for constructing algorithms capable of **learning.**



Definition by Tom M. Mitchell:

A computer program learns from experience **E** with respect to some class of problems **T** and a measure of quality **P** if the quality of solving problems from **T**, measured on the basis of **P,** improves with experience **E**.

Experience E - data and labels of the target feature (if any).

The problem class T determines what exactly is required to be obtained.

A machine learning algorithm is a specific set of actions to maximize the quality of problem solving by an algorithm for new use cases. Depending on the presence of a target feature, machine learning algorithms can be divided into those that are trained "with a teacher" and "without a teacher" **(supervised & unsupervised learning).**

Quality metric P is a numerical evaluation of the result of the algorithm.

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Bayes Theorem and Machine Learning

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Bayes formula follows from the definition of conditional probability. The probability of a joint event ***AB*** is expressed in terms of conditional probabilities:

$$p(ab) = p(a|b)p(b) = p(b|a)p(a)$$

where:
$\boldsymbol{p(h)}$— the probability that an event h occurs,
$\boldsymbol{p(ab)}$— the probability that an event a AND event b occure
$\boldsymbol{p(a|b)}$— the probability that a happened given that b happened.

In machine learning:
data - $\boldsymbol{D}$, hypotheses - $\boldsymbol{\theta}$ (parameters of the model).
The probability of the hypothesis for the data $\boldsymbol{p(\theta|D)}$, which, according to Bayes's theorem, is equal to:

$$\boldsymbol{p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)}}$$

$\boldsymbol{p(\theta)}$ – a priori probability
$\boldsymbol{p(D|\theta)}$ – likelihood
$\boldsymbol{p(D)}$ – constant
$\boldsymbol{p(\theta|D)}$ – a posteriori probability

We need to find a **hypothesis** $\boldsymbol{\theta}$ for which **a posteriori probability** $\boldsymbol{p(\theta|D)}$ is maximized. That is, for each hypothesis, calculate its **posterior probability** and choose the hypothesis for which this probability is maximized.
This is the **maximum a posteriori probability (MAP).**

$$\hat{\boldsymbol{\theta}}_{MAP} = \text{argmax}_{\boldsymbol{\theta}}\, p(D|\boldsymbol{\theta})p(\boldsymbol{\theta})$$

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https://machinelearningmastery.com/bayes-theorem-for-machine-learning/

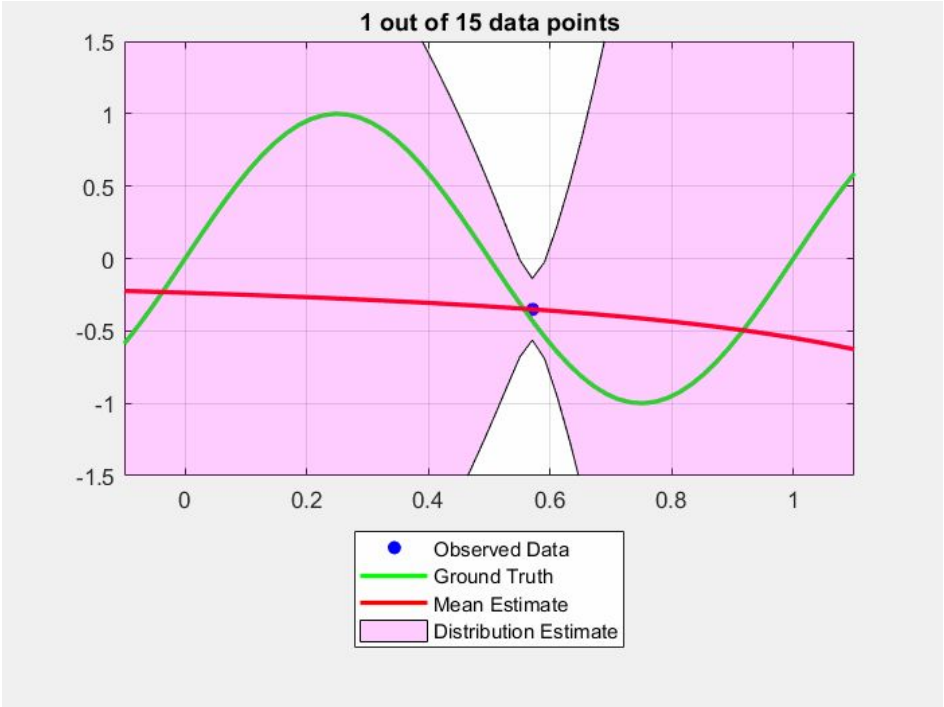
A posteriori probabililty maximization

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$$p(\boldsymbol{\theta}|D) = \frac{p(D|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(D)}$$

We need to find **hypothesis $\boldsymbol{\theta}$** which **likelihood is maximized**. That is, for each hypothesis we calculate its **posterior probability** and choose the one for which this probability is maximized. This is **maximum aposteriori probability (MAP)**:

$$\widehat{\boldsymbol{\theta}}_{MAP} = \text{argmax}_{\boldsymbol{\theta}}\, p(D|\boldsymbol{\theta})p(\boldsymbol{\theta})$$



Data, features, labels etc.

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```
df = pd.read_csv('assets/train.csv')
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

object

label

features

dataset = MATRIX [objects x features]

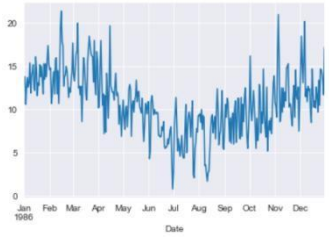
Data vizualization - matters!

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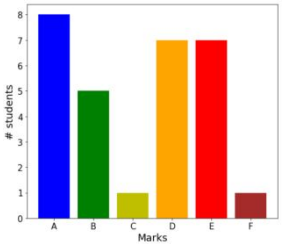
Matplotlib is a standard **Python** data visualization library;
The **matplotlib.pyplot** module provides a high-level library interface for plotting various types of plots.

The **Seaborn** library is an installable matplotlib add-on that provides tools for more attractive graph display;

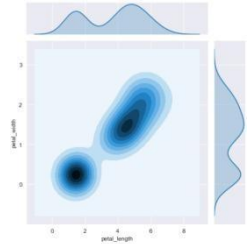
- convenient work with objects of the pandas package;
- building new types of graphic objects.



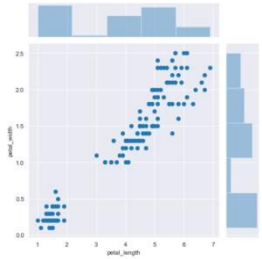
1D curve (x-y)



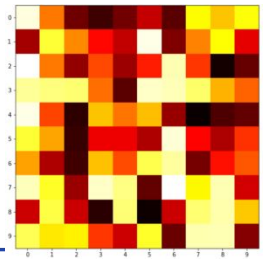
histogram



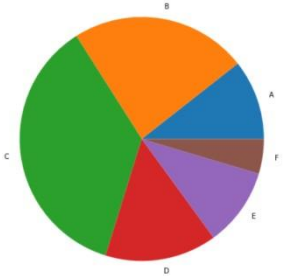
2D clouds with projections



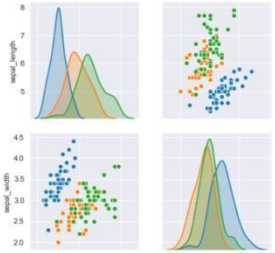
scatter-plot+ histogram



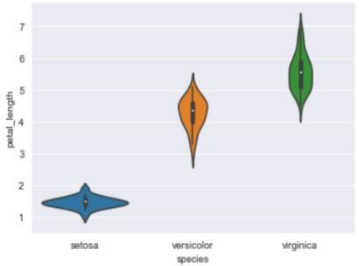
heatmap



pieplot



scatter-plot-matrix



violin-plot

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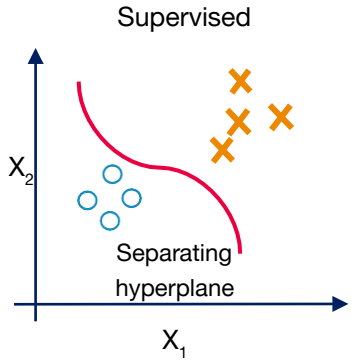
Supervised and unsupervised learning

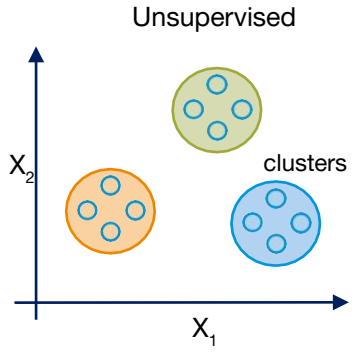
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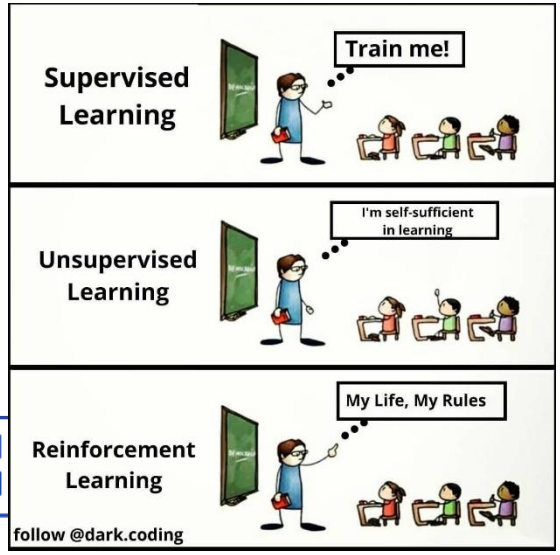
Supervised Learning - a machine learning approach where an algorithm is trained using labeled data, meaning each input is paired with a known, correct output

Regression - the output should be **a real number**.

Classification - the output should be **categorical (class)**







Unsupervised learning - no answers are given, and you need to look for dependencies between objects.

Clustering is the distribution of data into groups.

Reinforcement learning – learning strategy from experience.

How to test AI ?



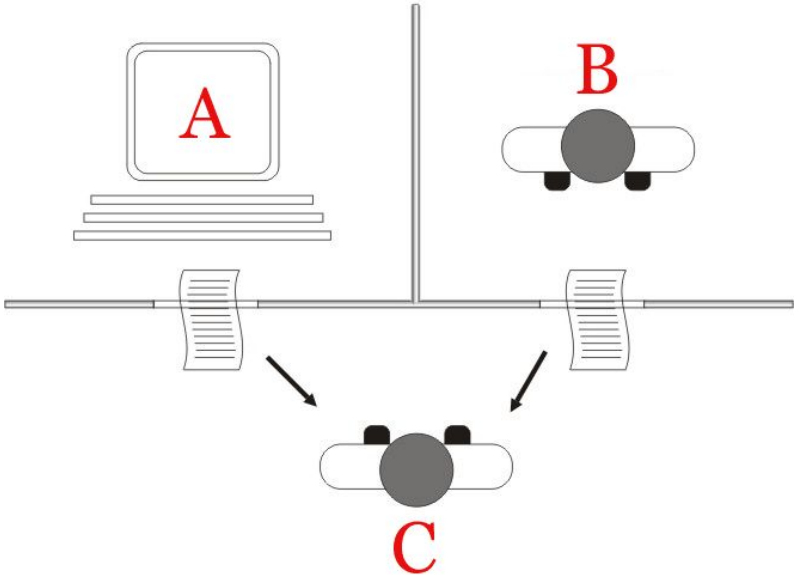
Alan Turing **[1912 - 1954]**

vs.



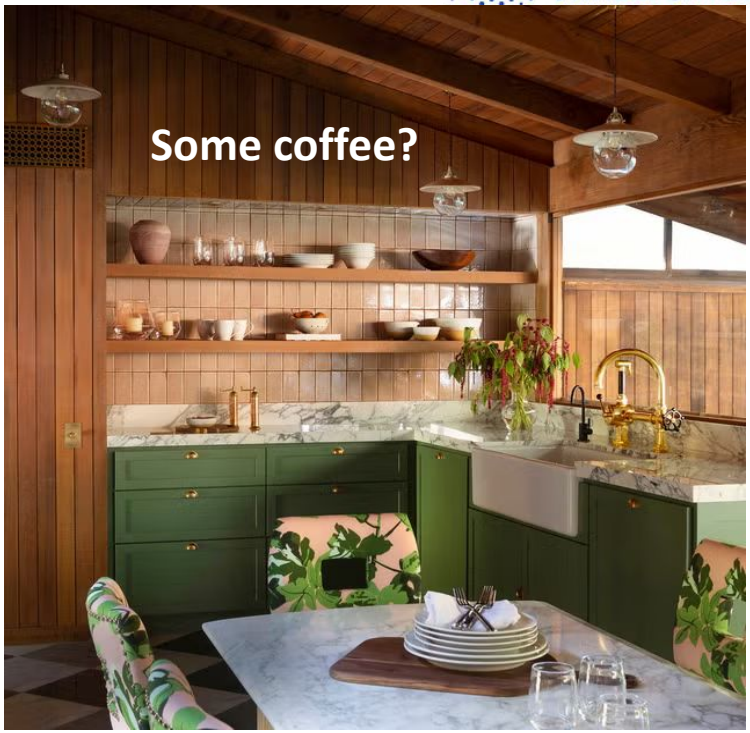
Steve Wozniak **(born 1950)**

How to test AI ?



Turing test

vs.



***Wozniak* test**

…to be continued

